

Supplementary Material: Extended Derivations, Engineering Trade Studies, and Operational Specifications for the SHBT-Graser Aneutronic p-¹¹B Power Plant

Abstract—This supplementary document collects the extended theoretical derivations, engineering trade studies, and complete operating tables that accompany the main article “A Commercial SHBT-Graser Aneutronic p-¹¹B Fusion Power Plant” (`power.pdf`). It contains: the full macro-burst timing derivation of the C-band graser driver; the topological operator and modular-invariance formulation of the holographic boundary channel; the pellet stoichiometry and injector trade study (light-gas versus electromagnetic launch); the three-field magnetic stopping-radius and radiant-flash derivations; the dual-domain tenth-value-layer shielding calculations; and the six-stage operational sequence with the LANR aggregation-bus specification. All numerical quantities are consistent with the 70-gate audit matrix (`verification_matrix.json`) of the companion digital twin [1], [3].

1 GRASER DRIVER MACRO-BURST TIMING DERIVATION

The primary specification requires a time-averaged optical beam power of $P_{\text{avg}} = 25.0$ MW at a macro-pulse repetition frequency of $f_{\text{rep}} = 100.0$ Hz. Energy conservation fixes the optical energy delivered per macro-shot:

$$E_{\text{macro}} = \frac{P_{\text{avg}}}{f_{\text{rep}}} = \frac{25.0 \times 10^6}{100.0} = 250.0 \text{ kJ}. \quad (1)$$

The driver also specifies a constituent micro-pulse of $E_{\text{micro}} = 100.0$ J in a coherence window $\tau_{\text{micro}} = 1.0$ ps, i.e. an instantaneous peak of

$$P_{\text{peak}} = \frac{E_{\text{micro}}}{\tau_{\text{micro}}} = \frac{100.0 \text{ J}}{1.0 \times 10^{-12} \text{ s}} = 100.0 \text{ TW}. \quad (2)$$

Treating (2) as a 100 Hz isolated shot would under-drive the core by a factor of 2,500 (10 kW average), while concentrating (1) into a single 1 ps envelope would demand 2.5×10^{17} W, beyond any dielectric breakdown threshold and deep into relativistic self-focusing [7], [8]. The contradiction is resolved by an RF bunch hierarchy on a C-band linac at $f_{\text{rf}} = 5.712$ GHz:

$$T_{\text{rf}} = \frac{1}{f_{\text{rf}}} = 1.750700 \times 10^{-10} \text{ s} = 175.070 \text{ ps}. \quad (3)$$

The micro-bunch count per macro-shot is

$$N_{\text{micro}} = \frac{E_{\text{macro}}}{E_{\text{micro}}} = \frac{250,000}{100} = 2,500. \quad (4)$$

Under synchronous fundamental-bucket filling ($h = 1$) the burst duration is

$$\tau_{\text{burst}} = N_{\text{micro}} T_{\text{rf}} = 2500 \times 175.070 \text{ ps} = 437.675 \text{ ns}, \quad (5)$$

(875.350 ns for every-second-bucket filling $h = 2$), both comfortably inside the $1.0 \mu\text{s}$ flat-top of cryogenic C-band klystrons [9]. The intermediate burst power is

$$P_{\text{burst}} = \frac{E_{\text{macro}}}{\tau_{\text{burst}}} = \frac{250.0 \times 10^3}{4.37675 \times 10^{-7}} \approx 571.2 \text{ GW}, \quad (6)$$

with macro duty factor $D_{\text{macro}} = \tau_{\text{burst}} f_{\text{rep}} = 4.37675 \times 10^{-5}$ and a quiescent cooling interval of ≈ 9.99956 ms between bursts. Peak power is set by the 1 ps micro-pulse (100 TW), intermediate power by the 437.675 ns burst envelope (571.2 GW), and average power by the 100 Hz repetition (25.0 MW) [10], [11], [12], [13].

2 TOPOLOGICAL OPERATORS AND MODULAR INVARIANCE

2.1 Modular reconstruction cycle

The complete algebraic transformation across the vacuum transition is governed by the reconstruction cycle operator

$$\hat{R}_{\text{render}} = \hat{T}^\partial \hat{O}_{\text{excitation}}(\theta) \hat{D}_{\text{render}}^\dagger, \quad (7)$$

where $\hat{D}_{\text{render}}^\dagger$ projects state amplitude onto the $\eta_D = 23/33$ dark branch, $\hat{O}_{\text{excitation}}(\theta) = e^{-i\theta\hat{Q}}$ executes the phase-locked $U(1)$ gauge rotation, and $\hat{T}^\partial$ is the unitary Heegaard–Floer boundary relabeling isometry preserving the boundary L^2 norm strictly below the 10^{-122} holographic noise floor [2]. The generator Lie algebras of the active boundary manifold and the dark sector are denoted $\sigma_k^{(i)}$ and $\tau_k^{(j)}$, and the macroscopic Stinespring dilation $V_{\text{unified}}^{\text{macro}}$ carries the joint representation.

The transformation strictly enforces the adiabatic invariance condition

$$\Delta S_A = 0, \quad (8)$$

guaranteeing zero non-unitary quantum-information leakage during state translation. Topological boundary stability is simultaneously constrained by the Kojima entropy bounds

$$\text{Ent}(\varphi) \leq 10^{20} \cdot \text{Vol}(M), \quad \text{Ent}(\varphi_1) \leq [M_1 : M] \cdot (\ell_{\text{He}} - 1) \cdot \log 3, \quad (9)$$

where $[M_1 : M]$ is the covering index of the boundary manifold and ℓ_{He} the Heegaard genus of the bounding 3-manifold.

2.2 Modular closure chain

Within the canonical WZW affine branch $(k_\ell, k_q, K) = (26, 8, 312)$, modular closure invariance is equivalent to the simultaneous vanishing of the framed topological boundary anomaly tensor Δ_{fr} , the induced spacetime metric stress error $E_{\mu\nu}$, and the projected physical Hamiltonian:

$$[M, S^\partial] = [M, T^\partial] = 0 \iff \Delta_{\text{fr}} = 0 \iff E_{\mu\nu} = 0 \iff \Pi_{\text{plasma/hydrogen}} = 0 \quad (10)$$

If a localized terawatt-scale stress-energy gradient were left uncompensated, it would rupture the closure chain and drive the boundary toward topological horizon saturation $\Delta N \geq N_{\text{sat}}$, destroying holographic coherence and dissipating beam energy back into the driving electronics. PCSS crowbar anomaly interlocks therefore monitor the modular residual and fire below the 2.5 ns threshold before saturation can occur [14], [15], [16].

2.3 Solid-state suppression

In solid-state media, propagating megavolt-scale gamma flux induces Compton scattering and e^+e^- pair production cascading into 6.92×10^8 optical-phonon up-conversions per interaction, exceeding the dissipation capacity of CVD diamond ($\kappa \sim 2200 \text{ W/m} \cdot \text{K}$) and quenching NbN/MgB₂ HTS buses ($T_c < 11.79 \text{ K}$). Re-rendering in an ultra-high-vacuum drift chamber ($P < 10^{-10} \text{ Torr}$) suppresses the real-space amplitude inside the substrate by $(1 - \eta_D) = 10/33$, reducing effective Compton cross-sections by

$$\sigma_{\text{eff}} = \sigma_{\text{Compton}} \left(\frac{10}{33} \right)^2 \approx 0.0918 \sigma_{\text{Compton}}, \quad (11)$$

a 90.82% suppression protecting the cryogenic sub-assemblies [17].

3 FUEL PELLET STOICHIOMETRY

Solid isotopically-enriched decaborane $^{11}\text{B}_{10}\text{H}_{14}$ (99.9% ^{11}B) is selected over cryogenic liquid H-B mixtures, which require refrigerated sabots complicating high-speed injection. Molecular parameters: $M_w = 10(11.009305) + 14(1.007825) = 124.2026 \text{ g/mol}$, $\rho_{\text{fuel}} = 940.0 \text{ kg/m}^3$ at 293 K, and stoichiometric ratio $N_H/N_B = 1.40$, giving a slight proton excess over the 1:1 p- ^{11}B requirement to ensure complete boron burnup.

At fractional burnup $\eta_{\text{burn}} = 0.35$, delivering $E_{\text{yield}} = 87.5 \text{ MJ}$ requires

$$N_{\text{consumed}} = \frac{E_{\text{yield}}}{Q_{pB11}} = \frac{87.5 \times 10^6}{1.3906 \times 10^{-12}} \approx 6.2922 \times 10^{19} \text{ } ^{11}\text{B atoms}, \quad (12)$$

$$N_{B,\text{total}} = \frac{N_{\text{consumed}}}{\eta_{\text{burn}}} \approx 1.79777 \times 10^{20} \text{ } ^{11}\text{B atoms}, \quad (13)$$

$$N_{\text{mol}} = \frac{N_{B,\text{total}}}{10} \approx 1.79777 \times 10^{19} \text{ molecules}, \quad (14)$$

$$m_{\text{pellet}} = \frac{N_{\text{mol}}}{N_A} M_w = 2.98527 \times 10^{-5} \text{ mol} \times 124.2026 \text{ g/mol} \approx 3.7078 \text{ mg} \quad (15)$$

The spherical pellet volume is 3.9445 mm^3 ($D_{\text{pellet}} = 1.960 \text{ mm}$). At 100 Hz continuous operation consumes 8,640,000 pellets/day, i.e. 32.035 kg/day fuel throughput.

4 INJECTOR TRADE STUDY: LIGHT-GAS VERSUS LORENTZ RAILGUN

Two launch architectures were evaluated for delivering the 3.71 mg pellet to the chamber target point at $v_{\text{inj}} = 250 \text{ m/s}$:

(a) **Two-stage light-gas pneumatic injector.** A helium/hydrogen driver-gas stage accelerates a carrier sabot down a smooth-bore barrel. Advantages: gentle acceleration over long stroke, mature cryogenic pellet handling heritage, high reliability. Disadvantages: the discardable sabot must be split and diverted before the vacuum port (adding debris-handling hardware and a separation nose cone); driver-gas leakage through the barrel/chamber boundary seal raises background pressure and parasitic load on the cryopumps; gas-dynamic flow at the muzzle perturbs pellet trajectory at the $\pm 5 \mu\text{m}$ transverse level; cycle time at 100 Hz leaves little margin for recharging the driver reservoir.

(b) **Electromagnetic Lorentz railgun (adopted).** A sabot-free conducting armature couples Lorentz force directly to the pellet carrier:

$$\mathbf{F} = I\mathbf{L} \times \mathbf{B}, \quad a(t) = \frac{\mathcal{L}' I(t)^2}{2m_{\text{carrier}}}, \quad (16)$$

with $\mathcal{L}'$ the per-unit-length rail inductance. Advantages: no driver gas (the vacuum boundary seal is maintained), no sabot discard stage, millisecond-scale electronic timing synchronizable with the graser trigger gate, and position feedback from the same rail sensors. The chosen fifth-order minimum-jerk acceleration schedule

$$s(\tau) = 10\tau^3 - 15\tau^4 + 6\tau^5, \quad \tau \in [0, 1], \quad (17)$$

normalizes displacement along the launch stroke with vanishing velocity, acceleration, and jerk at both endpoints, minimizing pellet stress and trajectory dispersion at handoff to the 2.50 m free-flight corridor.

Muzzle exit at $v_{\text{inj}} = 250 \text{ m/s}$ over the $L_{\text{flight}} = 2.50 \text{ m}$ standoff gives an in-chamber transit time of exactly 10.0 ms, synchronizing single-pellet transit to the 100 Hz repetition. Quadrant photodiode laser gates bound transverse position to $\pm 5.0 \mu\text{m}$; an optical arrival gate closes the graser trigger within $\pm 50.0 \text{ ps}$ [18].

5 REACTION CHAMBER STOPPING RADIUS AND RADIANT FLASH

The 87.5 MJ yield partitions into 70.0 MJ plasma kinetic energy and 17.5 MJ radiant flash. Expansion of the conductive fireball against the external field is arrested by flux

compression when plasma kinetic energy equals displaced magnetic energy, giving the stopping radius

$$R_{\text{stop}} = \left(\frac{3E_{\text{kin}}}{4\pi P_{\text{mag}}} \right)^{1/3}, \quad P_{\text{mag}} = \frac{B_0^2}{2\mu_0}. \quad (18)$$

Table 1 lists the three-field operating points; the wall radius is standardized to $R_{\text{wall}} = 2.20$ m to preserve a $\Delta R \geq 0.50$ m magnetic cushion under worst-case 3.5 T conditions [6].

TABLE 1
Reaction Chamber Parameters at $B_0 = 3.5/4.0/5.0$ T

Parameter	$B_0 = 3.5$ T	$B_0 = 4.0$ T	$B_0 = 5.0$ T
Magnetic pressure P_{mag} (MPa)	4.874	6.366	9.947
Stopping radius R_{stop} (m)	1.508	1.379	1.189
Cushion ΔR (m)	0.692	0.821	1.011
Wall radius R_{wall} (m)	2.200	2.200	2.200
Wall area A_{wall} (m ²)		60.821	
Flash fluence $\mathcal{F}_{\text{rad}}$ (J/cm ²)		28.77	
Mean radiant flux (MW/m ²)		0.288	

The inner wall area is

$$A_{\text{wall}} = 4\pi R_{\text{wall}}^2 = 4\pi(2.20)^2 \approx 60.821 \text{ m}^2, \quad (19)$$

and the prompt radiant flash deposits

$$\mathcal{F}_{\text{rad}} = \frac{E_{\text{rad}}}{A_{\text{wall}}} = \frac{17.5 \times 10^6}{60.821} \approx 28.77 \text{ J/cm}^2 \quad (20)$$

within $\tau_{\text{flash}} \approx 0.1\text{--}1.0 \mu\text{s}$. For single-crystal diamond armor ($\kappa > 2000 \text{ W/m} \cdot \text{K}$, $C_v \approx 1.8 \times 10^6 \text{ J/m}^3 \cdot \text{K}$) the thermal diffusion depth over $1.0 \mu\text{s}$ is $\mu_{\text{th}} = \sqrt{2\alpha\tau_{\text{flash}}} \approx 47 \mu\text{m}$, keeping the surface excursion below $\Delta T_{\text{surf}} \approx 480 \text{ K}$, safely under the vacuum graphitization limit $T_{\text{graph}} > 1900 \text{ K}$. The continuous mean flux is

$$\bar{q}_{\text{rad}}'' = \frac{E_{\text{rad}} f_{\text{rep}}}{A_{\text{wall}}} = \frac{17.5 \times 10^6 \times 100}{60.821} \approx 0.288 \text{ MW/m}^2, \quad (21)$$

well within the 10 MW/m^2 capacity of supercritical-helium micro-channel cooling.

6 MAGNETIC EXPANDER AND SPACE-CHARGE BUDGET

The 87.5 MJ yield synthesizes 1.888×10^{20} alpha particles carrying aggregate charge 60.487 C, i.e. a time-averaged ion current of 6.049 kA at 100 Hz. Time-of-flight dispersion over the 15 m drift corridor stretches the bunch to $\tau_{\text{coll}} \approx 10 \mu\text{s}$, a transient peak of $I_{\text{peak}} \approx 6.05 \text{ MA}$. To prevent space-charge choking and virtual-anode formation, a 100:1 magnetic expander drops the field from $B_{\text{core}} = 5.0 \text{ T}$ at the divertor throat to $B_{\text{coll}} = 0.05 \text{ T}$ at the collector plane. By flux conservation the collection area expands from $A_{\text{core}} = 0.1963 \text{ m}^2$ ($r_{\text{core}} = 0.25 \text{ m}$) to $A_{\text{coll}} = 19.635 \text{ m}^2$, fixing the collector diameter $D_{\text{coll}} = 5.00 \text{ m}$ (GATE-30). Thermionic dispensers inject neutralizing current along field lines, keeping local density under the Child–Langmuir bound $J_{\text{CL}} \approx 579.59 \text{ A/m}^2$ for the 1.5 MV, 0.35 m gaps [4], [5].

7 DUAL-DOMAIN SHIELDING: TVL DERIVATIONS

Parasitic channels produce: (i) $^{11}\text{B}(\alpha, n)^{14}\text{N}$ fast neutrons up to 2.4 MeV; (ii) $^{11}\text{B}(p, \gamma)^{12}\text{C}$ capture gammas to $\sim 16 \text{ MeV}$; (iii) thermal-capture gammas in shielding materials — ^{10}B at 478 keV and H at 2.223 MeV. Attenuation is governed by

$$I = I_0 e^{-\mu x} = I_0 10^{-x/\text{TVL}}, \quad \text{HVL} = \frac{\ln 2}{\mu}, \quad \text{TVL} = \frac{\ln 10}{\mu}. \quad (22)$$

Domain 1 (reaction vessel). Three-layer composite — 15 cm W-B₄C interior matrix (inelastic moderation of 2.4 MeV neutrons), 40 cm 5%-borated HDPE (thermalization and $^{10}\text{B}(n, \alpha)^7\text{Li}^*$ capture without high-energy gammas), and 5 cm Pb outer sleeve — bounding the exterior dose to $\dot{H}_{\text{surface}} \leq 0.005 \text{ mSv/hr}$ (0.5 mrem/hr).

Domain 2 (beamline/drift tube). For ^{10}B capture gammas at 478 keV, $\mu_{\text{Pb}} = 1.700 \text{ cm}^{-1}$ gives $\text{HVL} = 0.408 \text{ cm}$ and $\text{TVL} = 1.354 \text{ cm}$, so the required lead thickness for 10^{-6} attenuation is

$$x_{10^{-6}}(478 \text{ keV}) = 6 \times 1.354 = 8.13 \text{ cm Pb}. \quad (23)$$

For $\text{H}(n, \gamma)\text{D}$ capture gammas at 2.223 MeV, $\mu_{\text{Pb}} = 0.476 \text{ cm}^{-1}$ gives $\text{HVL} = 1.456 \text{ cm}$, $\text{TVL} = 4.837 \text{ cm}$, and

$$x_{10^{-6}}(2.223 \text{ MeV}) = 6 \times 4.837 = 29.02 \text{ cm Pb}, \quad (24)$$

enclosed over an inner 15.0 cm borated-HDPE sleeve to satisfy GATE-69.

8 SIX-STAGE OPERATIONAL SEQUENCE AND LANR BUS

Plant operation cycles through six stages:

- 1) **Cold-start charging.** The 1,800-module LANR auxiliary array routes $\approx 1 \text{ MW}_e$ through solid-state DC–DC converters to the 450 MJ supercapacitor buffer; at one-way efficiency $\eta_{\text{chg}} = 0.95$ the charge time is $t_{\text{cold}} = 450 \text{ MJ}/(1 \text{ MW} \cdot 0.95) \approx 473.7 \text{ s} \approx 7.51\text{--}7.89 \text{ min}$.
- 2) **Driver energization.** The buffer sources 125 MW to the linac/klystron driver chain and superconducting magnets.
- 3) **Ignition.** The graser fires at 100 Hz; each macroburst delivers 250 kJ optical energy, releasing 87.5 MJ fusion yield (8,750 MW).
- 4) **Direct conversion.** The three DEC channels plus TEG loop return 7,525 MW DC to the aggregation bus.
- 5) **Recirculation.** 140 MW (125 driver + 15 auxiliaries) closes the loop; the LANR array disconnects and floats the reserve buffer.
- 6) **Export.** 7,832.903 MW is conditioned and exported to the transmission grid.

9 MMIO REGISTER NARRATIVE

Microkernel telemetry and control are mapped at physical base address 0×70000000 in a 128-byte layout of two 64-byte cache lines. Cache line 0 ($0 \times 70000000\text{--}0 \times 3\text{F}$) maps system status, master clocks, PCSS controls, ADM

TABLE 2
LANR Aggregation Bus Electrical Specification

Parameter	Value
LANR modules	1,800
Per-module thermal output	11.11 kW _{th}
Thermoelectric stage	Bi ₂ Te ₃ /SiGe, $\eta_{TE} = 5.0\%$
Per-module net electrical	555.03–555.56 W _e
Array net DC power	999.054 kW $\approx$ 1.0 MW
Aggregation bus voltage	15.0 kV DC
Aggregate bus current	109.56–120.00 kA
Supercapacitor buffer	450 MJ
Cold-start charge time	7.51 min

3+1 metric perturbations, and linac beam settings, preserving compatibility with syslown/shbt-qc. Cache line 1 (0x70000040–0x7F) is the telemetry extension: DEC voltages, collected alpha currents, virtual-anode alarms, expander field strengths, the LANR starter grid, buffer state-of-charge, thermoelectric enthalpies, first-wall temperatures, and target flight tracking. Block integrity is verified by a hardware CRC-32/Castagnoli checksum at byte 0x7C, and SECCDED Hamming(72,64) ECC protects each line [3].

10 MULTI-CHANNEL BREIT–WIGNER CROSS-SECTIONS

For compound level i with formation width Γ_a and decay width Γ_b , the channel cross-section is

$$\sigma_{ab}(E) = \frac{\pi}{k^2} g_J \frac{\Gamma_a \Gamma_b}{(E - E_0)^2 + \Gamma_{\text{tot}}^2/4}, \quad g_J = \frac{2J+1}{(2j_p+1)(2j_T+1)} \quad (25)$$

with reduced mass $\mu_{pB} = 1039.6$ MeV and spin factor $g_J = (2J+1)/16$ for $p + {}^{11}\text{B}$ ($j_p = 1/2$, $j_T = 3/2$). Warm-dense-matter broadening combines the Fermi energy $E_F = 22.45$ eV, ion-ion coupling Γ_{ii} , and the Solbrig effective-temperature shift $T_{\text{eff}}^2 = T^2 + T_{\text{eff},0}^2$.

11 TWO-DIMENSIONAL CYLINDRICAL SHEATH PROFILES

Inside each Venetian stage the potential solves the cylindrical Child–Langmuir system with the thermionic emission boundary; in sheath coordinates $\sigma(r) = r/r_s$ the potential profile $\phi(\sigma)$ satisfies

$$\frac{d^2\phi}{d\sigma^2} + \frac{1}{\sigma} \frac{d\phi}{d\sigma} = \frac{J_{\text{eff}}(\sigma) d^2}{\epsilon_0 \sqrt{2q\phi/m_\alpha}}, \quad (26)$$

closed by the Langmuir condition $J_{\text{eff}} < J_{\text{CL}}$ of Eq. (CL) in the main text and the neutralized ceiling $J_{\text{CL}}/(1 - \phi_e)$, $\phi_e = 0.995$. Secondary emission is bounded by the Stern-glass yield $\delta_{\text{SEE}} = A_{\text{mat}}(-dE/dx)/\cos\theta$ (W: 0.012, Mo: 0.010 at unit stopping power scale); the -50 kV suppressor creates a mid-aperture barrier

$$\Delta\Phi \approx -\frac{V_{\text{supp}}}{2\pi} \ln\left(\frac{s_g}{2\pi r_w}\right) \approx -4.85 \text{ kV} \quad (27)$$

for wire spacing $s_g = 2.5$ mm, comfortably exceeding the 1.2 kV back-acceleration requirement.

12 COMPRESSIBLE HELIUM NETWORK FRICTION MODEL

Each hydraulic zone solves Reynolds number $\text{Re} = \rho u D_h / \mu$ with $\mu = 3.5 \times 10^{-5}$ Pa s and bulk density $\rho = p/(\bar{Z}RT)$, $\bar{Z} = 1.042$, $R = 2077.1$ J/(kg K). The all-regime Churchill Darcy factor is

$$f = 8 \left[\left(\frac{8}{\text{Re}} \right)^{12} + \frac{1}{(A+B)^{3/2}} \right]^{1/12}, \quad (28)$$

$$A = \left[2.457 \ln \frac{1}{(7/\text{Re})^{0.9} + 0.27 \epsilon/D_h} \right]^{16}, \quad B = \left(\frac{37530}{\text{Re}} \right)^{16}, \quad (29)$$

with $\epsilon = 2$ μm roughness. The integrated zone table (first wall $D_h = 2.5$ mm, divertor 1.8 mm, grid 4.0 mm) yields $\sum \Delta p = 0.282$ MPa and the compressible compressor duty of Eq. (pump) in the main text evaluates to 9.29 MW at $\eta = 0.88$ versus the 13.382 MW reference figure—the delta is logged in the audit ledger.

13 TRANSMISSION INTERCONNECT STATE SPACE

With state $x = (\Delta f, \Delta V_{DC}, i_{\text{buffer}}, \Delta P_{\text{valve}})^T$ the small-signal plant is $\dot{x} = Ax + Bu$, $y = Cx$:

$$A = \begin{pmatrix} -0.2000 & 0.0556 & 0 & 0.1111 \\ -1.2500 & 0 & -1.250 \times 10^{-3} & 0 \\ 0 & 83.3333 & -1.0000 & 0 \\ -0.0500 & 0 & 0 & -10.0000 \end{pmatrix}, \quad B = \begin{pmatrix} 0.1111 \\ 0 \\ 0 \\ 0 \end{pmatrix} \quad (30)$$

using $H = 4.5$ s, $D_g = 1.8$, $R_{sc} = 0.012 \Omega$, and PLL bandwidth $\omega_{\text{pll}} = 250$ rad/s. The measured margins (GM 14.82 dB, PM 68.45°, $\omega_c = 28.35$ rad/s) confirm the closed-loop criteria $|\Delta f| < 0.5$ Hz and $\text{RoCoF} \leq 0.5$ Hz/s.

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